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Paleomagnetism of the Oligocene Kangtuo Formation red beds (Central Tibet): Inclination shallowing and tectonic implications

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ABSTRACT

A paleomagnetic study on the red beds of the Oligocene Kangtuo Formation (Fm) was carried out in the Gerze Basin of the Lhasa terrane. A total of 700 samples were collected from 37 sites. Stepwise thermal demagnetization revealed that the main magnetic carrier is hematite. The natural remnant magnetization (NRM) consists of two components. A low-temperature component (LTC) is identified below 300 °C and is interpreted to be a recent viscous overprint, whereas a high temperature component (HTC) unblocks at ~665–690 °C and is interpreted to be the primary magnetization. The HTC distributions show a clear east–west elongated distribution, which is considered as reflecting inclination flattening of deposited magnetic remanence carriers. After inclination calibration using the E/I method, the HTC could pass both a reversal test and a fold test at 95% confidence level, showing the mean direction at D_s = 340.3°, I_s = 44.2°, with k = 63.0, and α_{95} = 3.1°, corresponding to a paleopole at 71.7°N, 339.3°E (A_{95} = 3.1°), and the paleolatitude of the sampling site at 25.9 ± 3.1°N. The paleolatitude is consistent with that expected from the coeval pole of the Qiangtang terrane obtained from volcanic rocks, suggesting that there has been no paleomagnetically-discernable latitudinal motion between the Qiantang and Lhasa terranes since ~30 Ma. Comparing our new data with the apparent polar wander paths (APWPs) of East Asian blocks (Cogné et al., 2013), Europe, and India (Besse and Courtillot, 2002), we have reached the following conclusions. (1) There is no significant paleolatitudinal difference observed between the Lhasa terrane and other central and northern Asian terranes at ~30 Ma. (2) The observed paleolatitude of the Kangtuo Fm is 8.0 ± 4.9° lower than the expected paleolatitude deduced from the data of stable Europe, highlighting the 'Asian inclination anomaly' phenomenon, but is 4.6 ± 5.1° higher than that deduced from the data of the India Plate, likely presenting a ~506 ± 561 km shortening between the India and Lhasa terrane since the Oligocene. It is inferred that this amount of shortening has been absorbed mainly by the Himalaya Orogenic Belt.

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1. Introduction

Available paleomagnetic data show that after the collision between the Himalayan terrane and the Eurasian plate, additional convergence of ~3000 km took place between the Indian Craton and Eurasian plate (Besse et al., 1984; Enkin et al., 1992), of which ~1800 km (Patzelt et al., 1996; Yi et al., 2011) was between the Indian Craton and the Lhasa terrane. According to the "fill-the-gap" solution, this shortening is interpreted as the evidence for the existence of the "Greater India" (Argand, 1924; Patzelt et al., 1996; Ali and Aitchison, 2005). At present, most parts of the Greater India have disappeared, and the timing of the shortening process has become a focus of studies. Intraplate shortening takes place often in the forms of folding and thrusting, which is quite common in the Himalayan region, for examples, the Main Central Thrust (MCT), the Main Boundary Thrust (MBT), and the Great Counter Thrust (GCT), etc. (DeCelles et al., 1998; Hodges, 2000; Yin, 2006) (Fig. 1). These major thrusts remained active after the Oligocene (Yin, 2006). Estimating the amount of the shortening quantitatively is of great importance for understanding the tectonic modeling for the evolution of this magnificent collision zone.

Paleomagnetic method can be used to determine the paleolatitude of a terrane in a certain geological time, and the relative posi-

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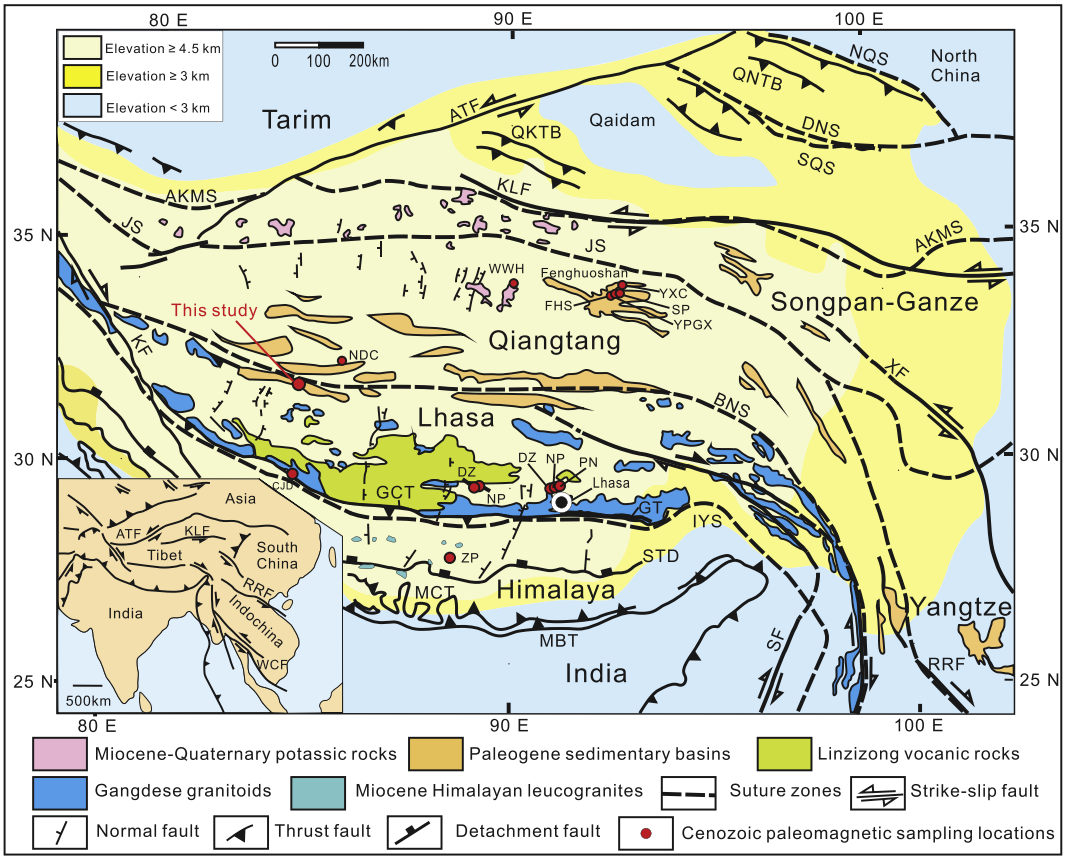


Fig. 1. Simplified geological map of the Tibetan Plateau and surrounding areas (modified from Chung et al., 2005; Yin and Harrison, 2000). Abbreviations: BNS, Bangong–Nujiang Suture; IYS, Indus–Yarlung Suture; JS, Jinsha Suture; KF, Karakoram Fault; GT, Gangdese Thrust; GCT, Great Counter Thrust; MCT, Main Central Thrust; MBT, Main Boundary Thrust; STD, South Tibet Detachment; AKMS, Anyemaqen–Kunlun–Muztagh Suture; ATF, Altyn Tagh Fault; KLF, Kunlun Fault; RRF, Red River Fault; SQS, South Qilian Suture; NQS, North Qilian Suture; DNS, Denghe Nanshan Suture; QKTB, Qiman Tagh-East Kunlun Thrust Belt; QNTB, Qilian Nanshan Thrust Belt; XF, Xianshuihe Fault; WCF, Wang Chao Fault; SF, Sagaing Fault. See in Table 2 for the abbreviation of the paleomagnetic investigation locations.

tions between tectonic blocks. For example, the latitude history of the Lhasa terrane can provide first-order information about the amount of India–Asia convergence that has been absorbed in Asia vs. India since continent–continent collision began (Besse et al., 1984; van Hinsbergen et al., 2011). In the Lhasa terrane, however, most of the available paleomagnetic data are from the Cretaceous and Eocene successions (Fig. 1) (e.g. Achache et al., 1984; Dupont-Nivet et al., 2010a; Tan et al., 2010; Chen et al., 2010, 2012; Meng et al., 2012; Huang et al., 2013; Ma et al., 2014; Yang et al., 2014). Paleomagnetic data from Middle Eocene and younger units from

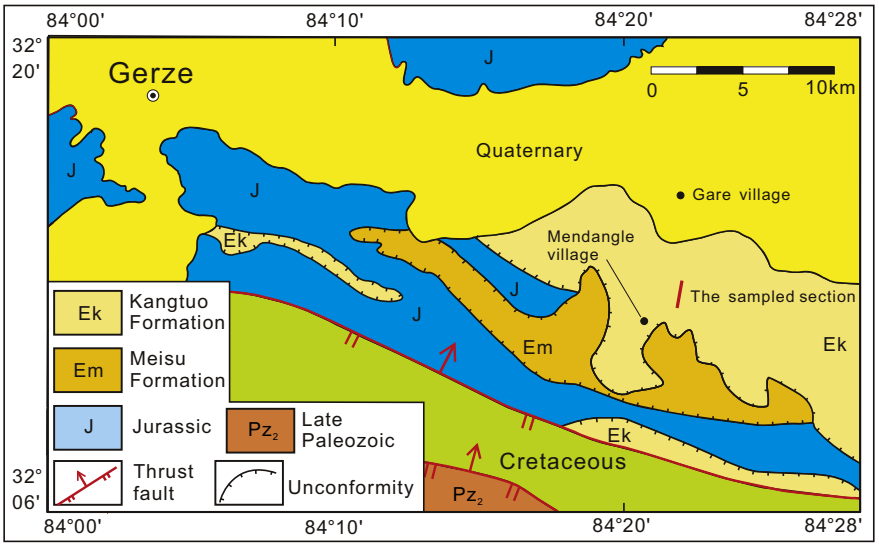


Fig. 2. Geological map of the study area and the location of the sampled section.

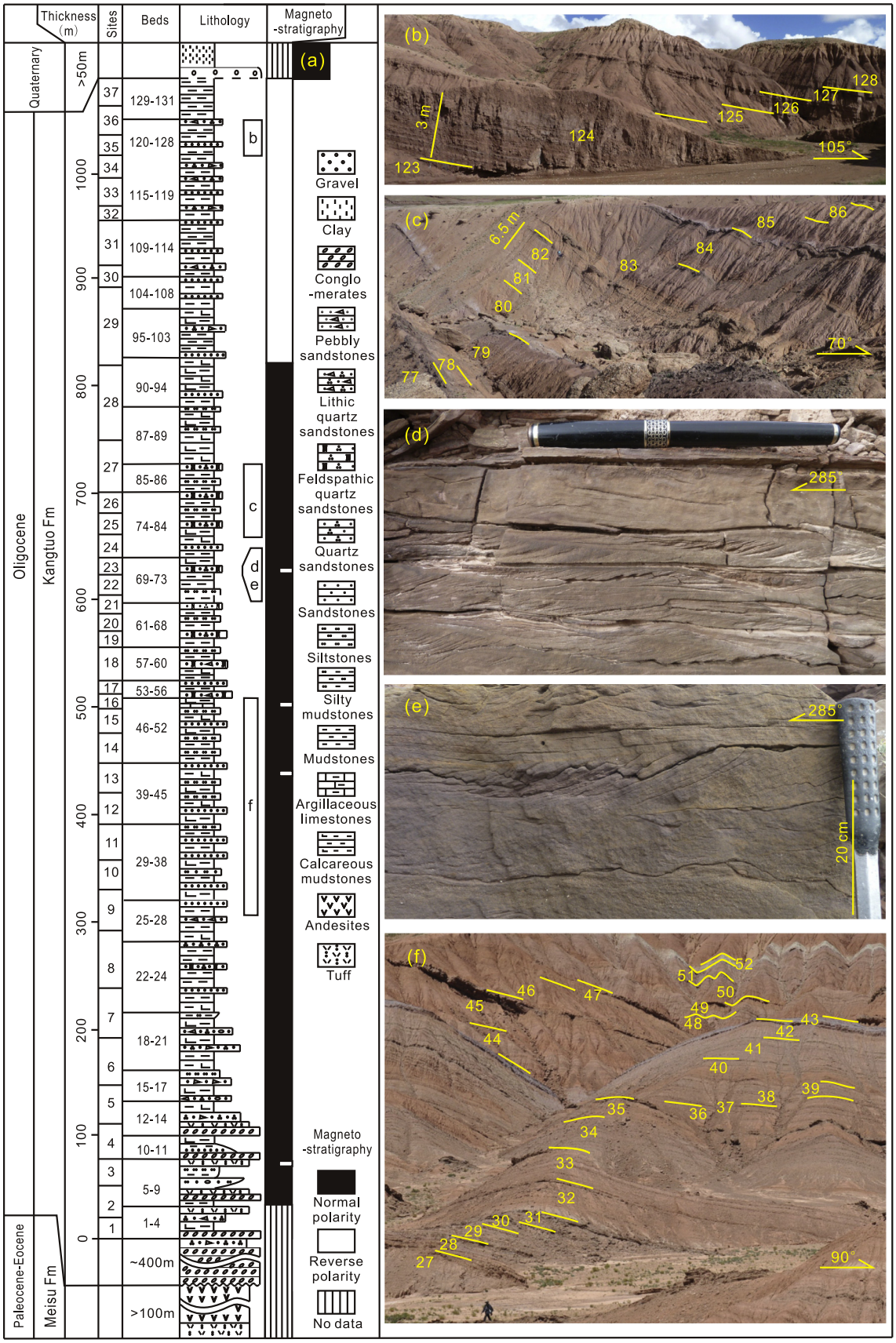


Fig. 3. (a) Composite stratigraphic column of the Gerze Basin (modified from Jiang et al., 2013). (b–f) Outcrop photographs, corresponding positions are marked in (a). (g) Variations of declination, inclination and virtual geomagnetic pole (VGP) latitude.

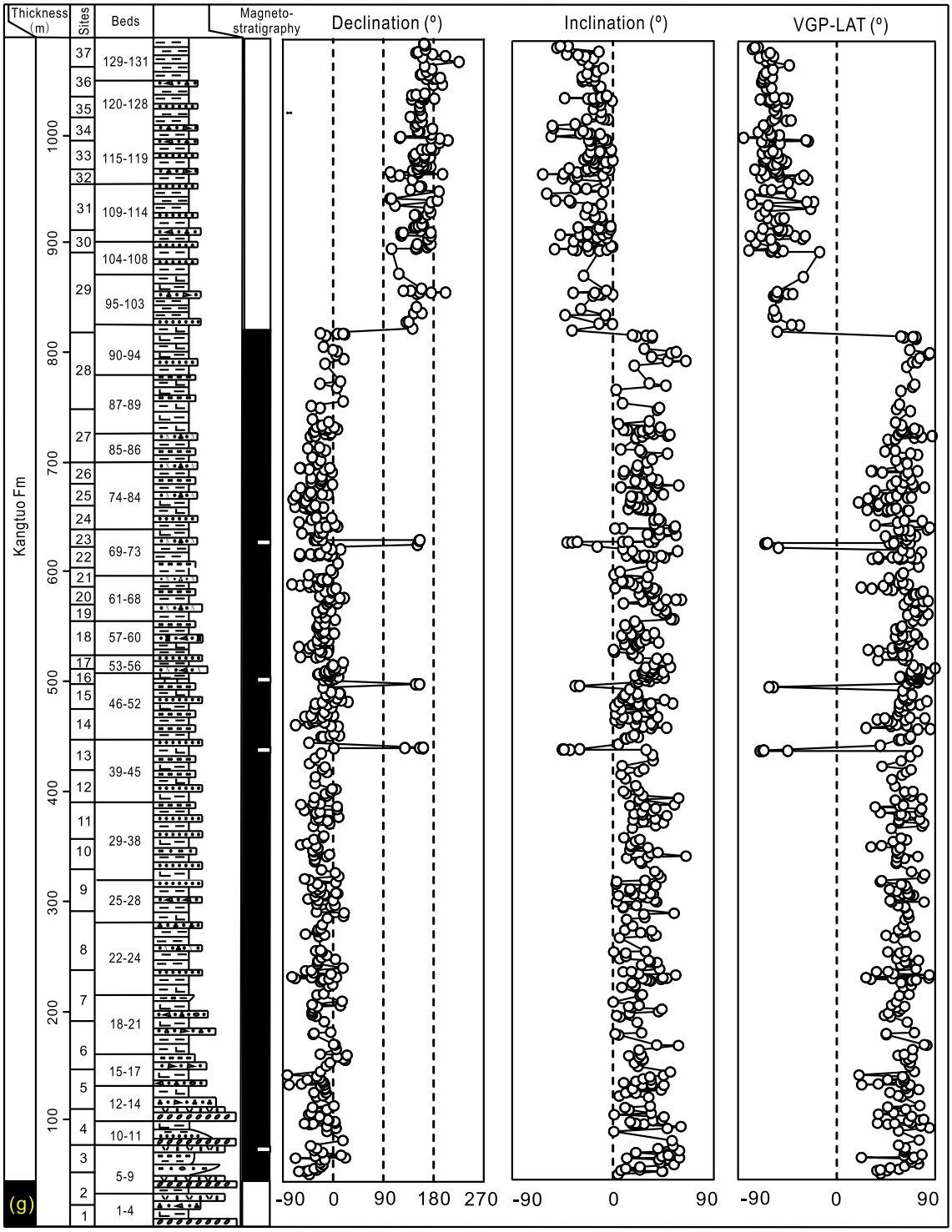


Fig. 3 (*continued*)

the Lhasa terrane are absent, and therefore we cannot estimate the shortening between the Lhasa terrane and the Indian craton since the Oligocene. To better understand the continental shortening after the India–Eurasia collision, more qualified Cenozoic paleomagnetic data are needed. In this study, new paleomagnetic data were obtained from the Oligocene Kangtuo Formation (Fm) of the Gerze Basin situated in the northern part of the Lhasa terrane, aimed to provide a tighter constraint on the tectonic shortening within the Qinghai–Tibetan Plateau and the adjacent regions since the Oligocene.

2. Geological setting

The sampled section of the Kangtuo Fm is at the Gerze Basin, ~30 km east of the Gerze County, near the northern margin of the Lhasa terrane, Tibet (Zhang et al., 2010, 2013). The basin preserves Paleogene sedimentary rocks, namely, from bottom to top, the Meisu Fm, the Kangtuo Fm, and Quaternary fluvial–lacustrine deposits (Zeng et al., 2006; Jiang et al., 2013). The Kangtuo Fm is wildly distributed and overlies the Triassic Wuga Fm, the Jurassic Mugagangriyan Fm, the Cretaceous Langshan Fm, and the Paleocene

to Eocene Meisu Fm (Figs. 2 and 3) (Zeng et al., 2006; Jiang et al., 2013). At the lowermost part (~400 m) of the Kangtuo Fm, it is dominated by conglomerate, which is not fit for paleomagnetic sampling. In the 1100 m thick sampling section (between 32.200°N/84.358°E and 32.214°N/84.366°E, Fig. 3), it consists mainly of lacustrine deposits, and was divided into 131 lithological beds (Jiang et al., 2013). The lower part of the section is dominated by purple thick-bedded (50–100 cm) conglomerate, reddish-brown thin-bedded (5–20 cm) lithic sandstone and mudstone. Three layers of tuffaceous rocks occur within the lower part of the section. The middle part of the sampled section contains reddish-brown medium- to thin-bedded mudstone (5–30 cm) intercalated with medium- to thin-bedded muddy siltstone and fine-grained sandstone, several cycles of muddy to sandy reversely graded bedding are observed throughout these intervals. The upper part is purple thick-bedded (50–100 cm) mudstone interbedded with medium- to thin-bedded (5–30 cm) siltstone (Fig. 3b). The Kangtuo Fm has been dated in separated basins in which volcanic rocks developed. In Zougouyouchacuo area (~100 km to the northeast of the Gerze Basin), a leucite phonolite bed in the upper Kangtuo Fm is dated of 30.5 ± 0.6 Ma ($^{40}Ar/^{39}Ar$, leucite, Li et al., 2006), and in Songxi area (~400 km to the northwest of the Gerze Basin), an andesite bed from the Kangtuo Fm was dated of 36.4 ± 0.9 Ma ($^{40}Ar/^{39}Ar$, biotite, Zhong et al., 2008). Based on these ages and regional stratigraphic correlation (Zhang et al., 2010, 2013), the age of Kangtuo Fm is assigned ranging 40–30 Ma in this study. This consideration is also consistent with the ages obtained from rocks that overlie this formation or are overlain by it. Such as, (1) in Bamaoqiongzong area (~400 km to the northeast of the Gerze Basin), the Yulinshan volcanic rock that unconformably overlie the Kangtuo Fm yields an age of 28.4 ± 1.3 Ma ($^{40}Ar/^{39}Ar$, biotite, Xu et al., 2013). (2) In Heihuling area (~400 km to the northeast of the Gerze Basin), the granite that is unconformably covered by the Kangtuo Fm was dated of 51.1 ± 1.5 Ma ($^{40}Ar/^{39}Ar$, biotite, Xu et al., 2013). (3) In the Nima area (~200 km to the east of the Gerze Basin), andesite and rhyolite beds of the Meisu Fm that is unconformably covered by the Kangtuo Fm were dated of 69.1–43.6 Ma (whole-rock, K–Ar, Xie et al., 2003).

Samples were cored in the field with a portable gas-powered drill. Cores were orientated with a magnetic compass and a sun compass. The differences between readings from the two compasses were within 2°. According to the stratigraphic cycling, 37 sampling sites were deployed in thickness intervals between 30

Table 1
Paleomagnetic results from the Oligocene Kangtuo Formation (reference point: 32.2°N, 84.3°E).

Site	Beds	n (N)	RStrike (°)	Dip (°)	D_g (°)	I_g (°)	D_s (°)	I_s (°)	k	α_{95} (°)	D_s^* (°)	I_s^* (°)	k^*	α_{95}^* (°)
1[R]	1–3	–	276	63	–	–	–	–	–	–	–	–	–	–
2[R]	4–6	1	276	63	293.0	38.5	317.3	4.5	–	–	317.3	8.9	–	–
3	7–9	17	312	57	284.6	33.1	330.5	38.4	12.0	10.7	328.8	54.5	13.4	10.1
4	10–12	17	299	56	286.8	51.3	342.2	33.1	11.3	11.1	341.5	48.9	11.3	11.1
5	13–16	27	296	70	272.1	50.1	343.1	30.4	10.8	8.9	342.5	45.8	12.8	8.8
6	17–19	13	284	65	291.3	54.2	336.9	16.0	13.8	11.6	340.2	34.4	9.6	12.1
7	20–22	19	292	43	298.6	53.1	337.0	32.6	12.1	10.1	336.6	49.6	14.7	9.1
8	23–25	15	281	65	285.1	63.2	342.3	20.4	16.7	9.6	342.2	34.1	13.3	10.9
9	26–29	20	298	51	303.7	46.4	339.4	23.8	13.6	9.2	340.1	38.2	10.9	10.4
10	30–32	11	312	41	304.7	41.8	338.3	34.4	17.0	11.4	337.7	51.6	20.2	10.4
11	33–38	20	289	43	300.5	51.9	335.0	29.4	16.1	8.4	333.9	46.3	18.3	7.8
12	39–41	12	295	52	307.0	49.8	342.3	21.3	17.2	10.8	342.2	35.4	15.1	11.5
13	42–45	14	295	46	307.1	53.8	344.1	28.1	15.5	10.4	344.3	44.6	14.6	10.8
14	46–49	19	294	44	306.7	42.4	334.2	21.1	14.4	9.2	334.3	34.5	10.9	10.6
15	50–51	11	314	50	327.0	50.1	1.4	22.5	16.9	11.4	1.1	36.1	13.7	12.8
16	52–53	12	310	42	314.0	47.5	348.1	31.1	39.2	7.0	348.3	49.7	50.5	6.2
17	54–56	18	308	35	320.2	53.5	352.2	35.9	23.6	7.3	352.0	54.5	36.2	5.8
18	57–60	16	295	45	316.4	38.1	336.5	18.0	23.2	7.8	336.1	31.5	16.8	9.3
19	61–64	11	301	36	316.6	48.4	343.4	30.0	19.3	10.7	343.3	46.1	17.7	11.2
20	65–66	13	305	44	297.9	59.1	352.2	41.4	24.1	8.6	351.9	58.6	26.5	8.2
21	67–69	17	295	45	307.1	45.6	336.4	23.4	11.8	10.8	336.9	38.2	10.3	11.7
22	70–71	11	301	42	309.0	48.2	337.6	33.7	12.1	13.7	337.3	51.3	16.9	11.4
23	72–73	14	301	42	307.4	43.9	337.3	27.7	33.3	7.0	337.4	44.3	22.1	8.6
24	74–77	18	295	38	305.4	44.6	329.3	32.6	12.9	10.0	329.5	49.5	13.8	9.7
25	78–80	14	290	35	298.3	42.9	323.4	29.8	12.4	11.8	323.3	46.6	14.0	11.0
26	81–84	20	292	43	302.7	44.2	330.6	26.4	20.8	7.3	330.5	43.2	19.9	7.5
27	85–88	18	293	56	285.7	56.2	343.3	32.2	22.4	7.5	343.9	49.5	19.8	8.0
28	89–93	16	305	45	316.8	62.9	2.3	34.3	14.1	10.2	2.2	50.5	13.6	10.4
29	94–107	31	305	45	126.5	−31.4	149.3	−20.7	10.6	8.3	149.8	−38.5	8.5	9.4
30	108–110	16	298	55	119.7	−37.2	150.6	−19.1	11.2	11.5	152.2	−36.6	11.1	11.6
31	111–114	15	304	27	130.9	−43.1	152.2	−34.6	10.0	12.8	152.8	−55.8	12.8	11.1
32	115	11	304	37	141.9	−31.5	158.0	−13.2	47.7	6.7	157.5	−28.6	20.5	10.3
33	116–118	20	288	33	151.3	−43.6	164.3	−18.0	11.5	10.0	164.1	−35.9	10.1	10.8
34	119–121	9	282	34	134.8	−48.9	154.4	−25.2	14.4	14.1	153.7	−43.3	11.2	14.5
35	122–125	15	310	22	154.0	−26.5	161.7	−15.1	21.9	8.4	162.0	−31.9	16.9	9.6
36	126–129	10	335	11	166.9	−23.3	171.6	−20.4	44.7	7.3	171.5	−41.9	39.0	7.8
37	130–131	10	30	28	163.3	−14.6	172.8	−33.6	11.5	14.9	175.0	−56.6	16.7	12.2
Mean	35						340.3	27.4	58.5	3.2	340.3	44.2	63.0	3.1
Corresponding pole							Plon 314.3	Plat 64.8	A_{95} 2.6		Plon* 339.3	Plat* 71.7	A_{95}^* 3.1	

Abbreviations: n (N), number of samples (sites) used to calculate the mean direction; RStrike, right hand strike of the bed; Dip, dip of the bed; D_g/I_g, declination/inclination in the geographic coordinates; D_s/I_s, declination/inclination in the stratigraphic coordinates; k, Fisher precision parameter of the mean; α_{95}/A_{95}, the radius of circle of 95% confidence about the direction/pole; Plat/Plon, latitude/longitude of VGP in stratigraphic coordinates. $D_s^*/I_s^*/k^*$/Plat*/Plon*, declination/inclination/Fisher precision parameter of the mean/latitude of VGP/longitude of VGP after E/I corrections; α_{95}^*/A_{95}^*, the radius of circle of 95% confidence about the direction/pole after E/I corrections;
Note: After the E/I corrections, (1) The McElhinny (1964) fold test is positive at 95% and 99% confidence levels: $k_s/k_g = 2.39 > F(2*(n2-1),(n1-1))$ at 5% and 1% point = 1.49 and 1.77. (2) The McFadden (1990) fold test is also positive at 95% and 99% confidence levels at "Xi1" test: critical Xi at 95% and 99% = 6.88 and = 9.73, respectively. "Xi1" IS = 13.73, "Xi1" TC = 3.69. (3) The classification of reversals tests is B (McFadden and McElhinny, 1990): mean angle = 4.1° < γ_c = 7.1°.
[R] Sites were not used to calculate the final mean direction.

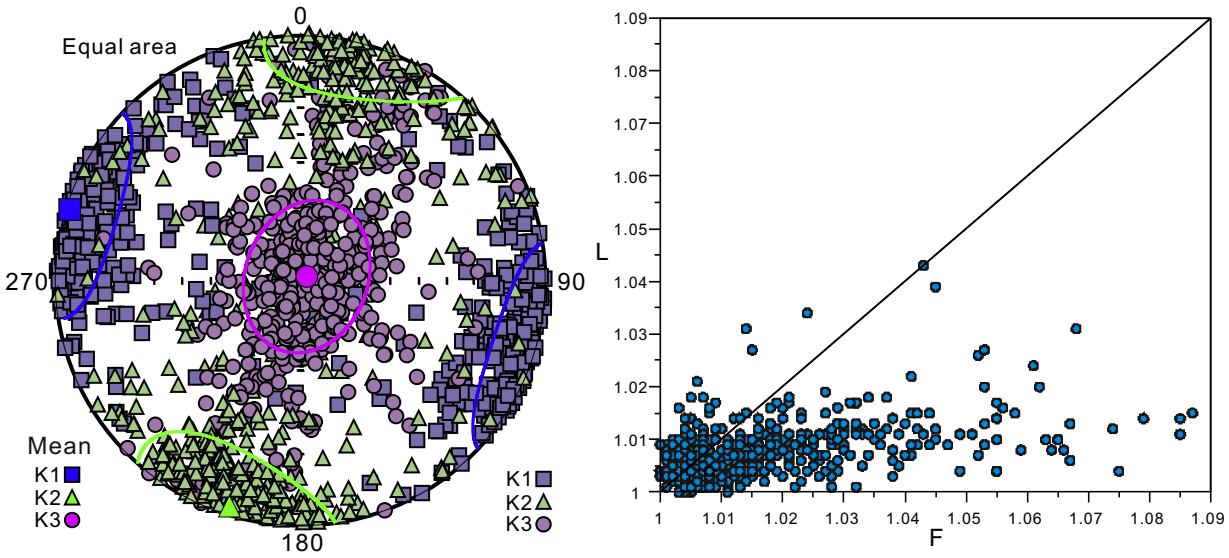


Fig. 4. Stereoplots (equal area, lower-hemisphere projection) showing AMS data from the Kangtuo Fm and the Plot of lineation vs. foliation. Maximum, intermediate, and minimum susceptibility principal axes are indicated, respectively, by triangles, squares and circles. $F = K2/K3$, $L = K1/K2$ (Jelinek, 1981).

and 50 m (Fig. 3a). The total number of orientated cored samples is 700 and the samples were re-shaped indoor into standard-sized specimens for paleomagnetic measurement (diameter 2.54 cm, height 2.2 cm).

3. Measurements and results

3.1. Laboratory procedures

All the measurements were performed in the Laboratory for Paleomagnetism and Environmental Magnetism, China University of Geosciences (Beijing). Specimens were first measured for magnetic fabric with a MFK1-FA magnetic susceptibility meter (AGICO, Czech). Specimens were stepwise thermally demagnetized in a TD-48 oven (ASC, USA) and magnetic remanence component was measured using a 2G-755-4K superconducting magnetometer. Remanence components were separated using the principal component analysis method of Kirschvink (1980), and remanence directions were analyzed using the Fisher statistics (Fisher, 1953).

3.2. Magnetic fabrics

Anisotropy of magnetic susceptibility (AMS) was determined for 503 specimens. The susceptibility is in $n \times 10^{-4}$ SI magnitude. The minimum principle susceptibility axes are in average perpendicular to the bedding plane, and the maximum axes are obviously concentrated in SEE–NWW direction (287 ± 26.2°) (Fig. 4) Our field measurements of the cross beddings and climbing beddings showed a flow direction of about 285° (Fig. 3d and e), which coincides well with the maximum axes of the AMS. The mean value of anisotropy degree (P) was quite small (1.025), magnetic foliation (1.017) is greater than the magnetic lineation (1.007) but all are very small in average, indicating that the AMS reflects typical sedimentary fabric and no significant tectonic stress has been applied to the rocks (Kodama, 2012).

3.3. Demagnetization results

Among the 700 specimens that underwent stepwise thermal demagnetization, 679 revealed useful magnetic signals, while the rest specimens did not due to coarse granularity, irregular sample shape, or were broken up during heating. The intensity of the natural remanent magnetization (NRM) in the specimens following complete demagnetization is generally between 1 and 10 mA/m (Fig. 5). The demagnetization characteristics were classified into three groups. In the first group of 346 specimens, a stable low-temperature component (LTC) and a stable high-temperature component (HTC) are observed. The second group, 205 specimens, revealed only a HTC with directions similar to that of the first group. The third group, 128 specimens, showed a LTC similar to that of the first group, while the HTC is unstable in direction. In total, 474 specimens presented LTC that was isolated before 300 °C. In geographic coordinate, the average LTC was $D_g = 359.4°$, $I_g = 49.1°$, $k = 23.5$, $\alpha_{95} = 1.4°$. The LTC directions distribute near the modern geomagnetic field ($D = 1.0°$, $I = 49.7°$) (Fig. 6). Therefore, it was interpreted as a viscous remanence obtained in the modern geomagnetic field.

The HTC was generally unblocked at 655–690 °C (Fig. 5), indicating that the magnetic carrier was hematite. The HTC deviates clearly from the modern geomagnetic field direction (Figs. 7 and 8). It included both normal and reverse polarities. Specimens from the lower part of the section in Beds 1–93 (Sites 1–28) showed mostly normal polarity, while the upper part, Beds 94–131 (Sites 29–37), in reverse polarity. Therefore, the HTC may be very meaningful for magnetostratigraphy (Fig. 3g, Table 1).

3.4. Inclination flattening and its correction

Previous studies showed that the magnetic inclination recorded in sedimentary rocks might have been significantly flattened (King, 1955; Lovlie and Torsvik, 1984; Jackson et al., 1991; Gilder et al., 2001, 2003; Tan et al., 2003, 2007). At present, applicable correction algorithms include anisotropy-based rock magnetism method (Kodama, 2009) and the E/I method (Tauxe and Kent, 2004; Tauxe et al., 2008). Numerous papers reported inclination flattening in the specimens from the Qinghai–Tibetan Plateau and the successful corrections using the E/I method (Yan et al., 2005; Dupont-Nivet et al., 2010b; Tan et al., 2010; Huang et al., 2013). It is obvious that our HTC distribution showed an east–west elongation (Fig. 7). Therefore, we inferred that the inclination in the Kangtuo Fm has been flattened. In the E/I method, the minimum number of specimens is 100, and we had on-hand 551 specimens, which meets the requirement.

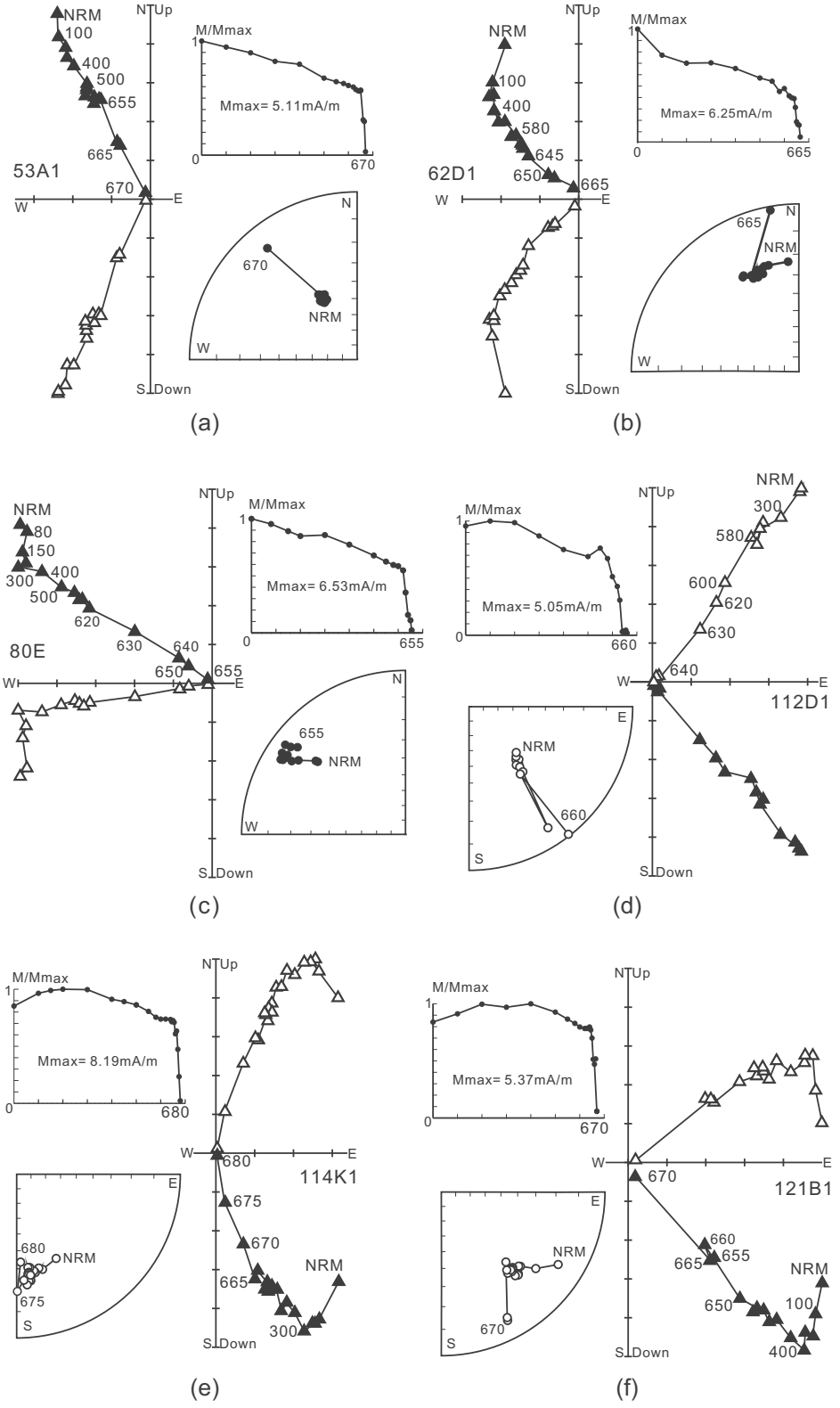


Fig. 5. Orthogonal vector plots showing the results of stepwise demagnetization of representative samples from the Kangtuo Fm, all in geographic coordinates. In orthogonal vector plots, closed (open) triangle symbols represent projections on the horizontal (vertical) plane. NRM, Natural Remanent Magnetization. In steroplots, closed (open) symbols represent positive (negative) inclinations.

The upper part of the section (Beds 94–131) is composed mainly of mudstone and the lower part (Beds 1–93) is more silty in lithology, and the HTC distribution shows that the direction distribution for the upper part of the section is more elongated than that for the lower part of the section. Therefore, the correction works were done in two fashions. In the first fashion, we do the

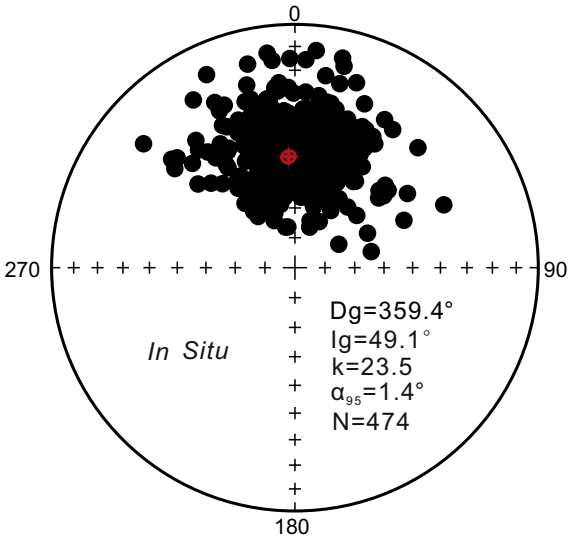


Fig. 6. Equal-area projections for the LTC of the Kangtuo Fm, in geographic coordinates. Star with oval indicates the mean direction with circle of 95% confidence.

same correction for all data from the whole section; in the second fashion, we do correction for those from the lower section (silty sample predominated) and the upper section (mudstone predominated), separately. In the first fashion, the flattening correction factor (f) was 0.5, and the inclination was corrected to 44.7° (95% confidence interval falls in 41.4–48.1°) (Fig. 7b, d, and e). The inclination-corrected average direction is $D_s = 339.1°$, $I_s = 42.7°$, $k = 12.0$, $\alpha_{95} = 1.8°$. In the second fashion, for 414 specimens in the lower section, $f = 0.5$, the inclination was corrected from 28.9° to 46.2°, 95% confidence interval falls in 42.5–50.0° (Fig. 7f and g). For the 137 specimens in the upper section, $f = 0.4$, the inclination is corrected from 21.1° to 42.6°, 95% confidence interval falls in 34.7–49.8° (Fig. 7h and i). The inclination-corrected average direction in Fashion 2 is: $D_s = 339.5°$, $I_s = 44.0°$, $k = 12.4$, $\alpha_{95} = 1.8°$, which is roughly the same to that of Fashion 1. Because the upper part of the succession contains more mudstones than the lower part does, fashion 2 reflects that the flatting is a function of grain size. This observation suggests that a universal flattening factor (e.g., $f = 0.6$) should be applied with caution (e.g., Torsvik et al., 2012). This is also consistent with the common paleomagnetic knowledge established since earlier time (King and Rees, 1966; Kodama, 2012; Huang et al., 2013), so we suggest that the Fashion 2 is more reasonable. The site mean directions of the corrected data

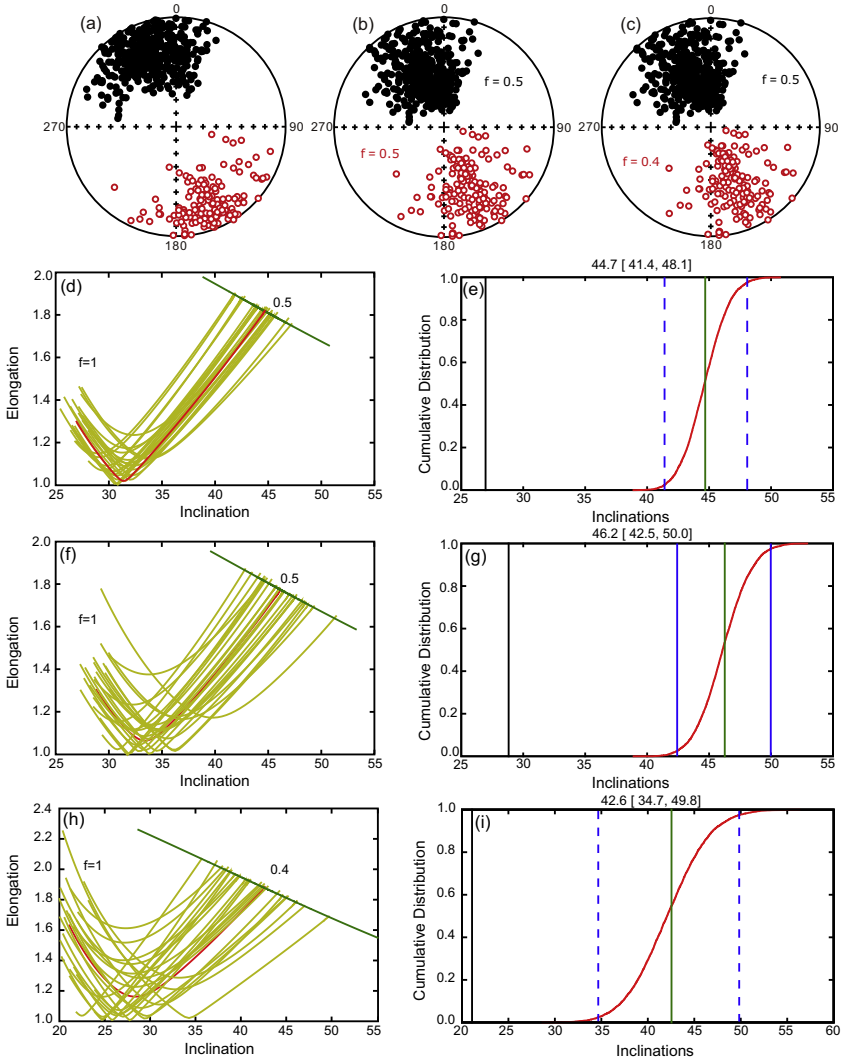


Fig. 7. Correction for the inclination error using the E/I method developed by Tauxe and Kent (2004). (a) Equal-area projections of the HTC directions before E/I correction. (b) Equal-area projections of the HTC directions after E/I correction by the Fashion 1. (c) Equal-area projections of the HTC directions after E/I correction by the Fashion 2. (d) Plot of elongation vs. inclination as a function of f on the Fashion 1. (e) Cumulative distribution of corrected inclinations on the Fashion 1. (f) Same as (d) but of the Beds 1–93 by the Fashion 2. (g) Same as (e) but of the Beds 1–93 by the Fashion 2. (h) Same as (f) but of the Beds 94–131. (i) Same as (g) but of the Beds 94–131.

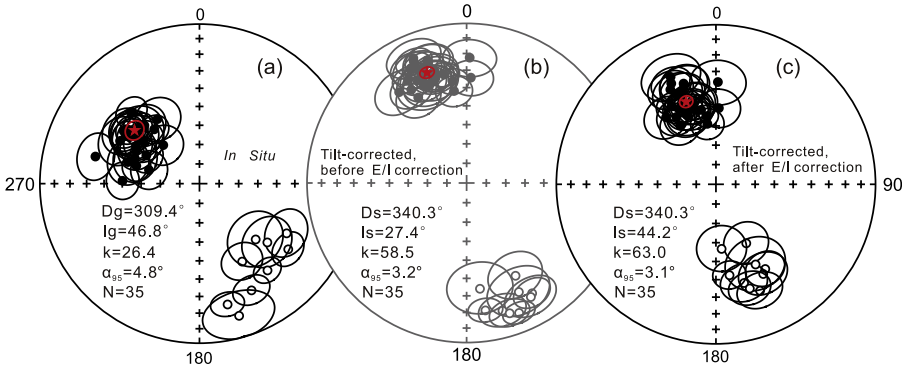


Fig. 8. Equal-area projections of the site-mean directions for the Kangtuo Formation before and after tilt correction. Solid (open) symbols represent positive (negative) inclinations; stars indicate the mean directions with 95% confidence limits of 35 sites.

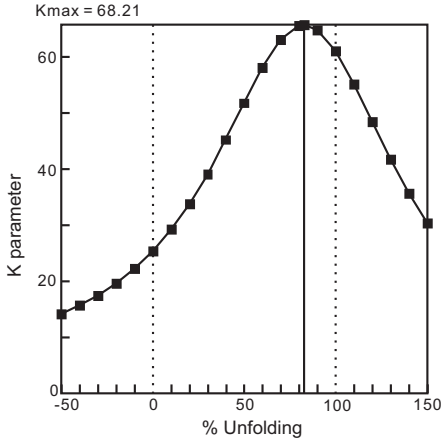


Fig. 9. Stepwise unfolding test of the mean direction of the HTC from the Kangtuo Fm.

could pass a class B reversal test: $\gamma_o = 4.1° < \gamma_c = 7.1°$ (McFadden and McElhinny, 1990), a fold test ($k_s/k_g = 2.39$), at 95% confidence level and the site mean directions yield a maximum grouping at 83% unfolding (Fig. 9) indicating the HTC acquired before the strata folded. The corresponding paleopole of the inclination-corrected HTC is at 71.7°N, 339.3°E, $A_{95} = 3.1°$, and the paleolatitude of the sampled section is 25.9 ± 3.1°N. This paleolatitude is consistent with that deduced from the coeval volcanic rocks (~35 Ma) from the Qiangtang terrane (Lippert et al., 2011), supporting the idea that the Qiangtang terrane and the Lhasa terrane can be regarded as a coherent tectonic unit in the Cenozoic (Kapp et al., 2005, 2007a, 2007b).

4. Discussion

Currently, available Paleocene-Eocene paleomagnetic data of Lhasa and Qiangtang terranes can be classified into two groups. Taking location (32.2°N, 84.3°E) in the Gerze Basin that situated in the northern Lhasa terrane as a reference point, data of the first group show a low-paleolatitude of 16.1 ± 2.1°N (A1 in Table 2), and

Table 2
Summary of Cenozoic paleopoles from the Lhasa and Qiangtang terranes.

Pole	Lithology	Slat/Slon (°N/°E)	Age (Ma)	Plat/Plon (°N/°E)	A_{95} (°)	P_λ (°N)	n (N)	Criterion (Q)	Reference
Lhasa terrane									
KTo*	Red beds	32.2/84.3	Oligocene	64.8/314.3	2.6	14.5	471 (35)		This study
KTc	Red beds	32.2/84.3	Oligocene	71.7/339.3	3.1	25.9	471 (35)	123F5R7 (7)	This study
DZ	volc	29.9/90.2	64–60	66.8/286.8	6.6	10.5	93 (15)	123F5□7 (6)	Chen et al. (2010)
NPC	sed	29.9/90.2	60–50	70.8/290.2	6.2	14.6	121 (18)	123F5D7 (7)	Chen et al. (2010)
NPS	volc	29.97/91.13	55.7 ± 3.4	73.4/281.7	9.2	16.3	58 (8)	123F5D7 (7)	Sun et al. (2010)
PNT	tuff	30.0/91.1	56–46	87.5/80.4	5.8	34.7	76 (9)	123F5□7 (6)	Tan et al. (2010)
PND	volc	30.0/91.1	56–46	76.4/212.7	5.2	23.2	195 (24)	123F5D7 (7)	Dupont-Nivet et al. (2010a)
PNL	volc	30.0/91.1	56–46	80.2/230.4	4.1	23.9	291 (41)	123F5D7 (7)	Lippert et al. (2014)
LIE	Dyke	30.0/91.1	52.9 ± 0.4	71.9/225.7	5.2	16.6	57 (9)	123□5D7 (6)	Liebke et al. (2010)
PNH	sed	30.0/91.1	43.5–42.3	78.2/215.7	2.6	24.0	119	123□5□7 (5)	Huang et al. (2013)
CJD	sed	29.9/84.3	57–54	78.0/329.0	6.8	26.5	62	123F5□7 (6)	Meng et al. (2012)
Qiangtang terrane									
QT	volc	33.1/87.5	35 ± 3.0		3.7	28.0	225 (20)	123□5R7 (6)	Lippert et al. (2011)
FHSL	Red beds	34.6/92.8	E_1-E_2	55/201	4.2	12.6	57 (12)	123□5D7 (6)	Lin and Watts (1988)
FHSH	Red beds	34.5/92.8	E_1-E_2	72.2/242.1	7.7	15.5	38 (7)	123F5R7 (7)	Halim et al. (1998)
YXC	Red beds	35.0/93.0	30.5 ± 0.5	55.7/192.7	6.3	16.8	59	123F5R7 (7)	Liu et al. (2003)
SP	Red beds	34.7/92.9	35.8–30.9	73.7/308.1	7.5	19.9	51	123F5D7 (7)	Liu et al. (2003)
GS	Red beds	34.7/92.9	39.6–35.8	69.1/207.7	8.4	19.4	82	123□5D7 (6)	Liu et al. (2003)
YPGX	Red beds	34.5/92.7	50.8–39.6	73.0/226.7	5.1	18.3	94	123F5D7 (7)	Liu et al. (2003)
Average									
A1 mean paleolatitude taking location of the Gerze Basin (DZ, NPC, NPS, FHSL, FHSH, YXC, SP, GS, YPGX, LIE)									16.1 ± 2.1°N
A2 mean paleolatitude taking location of the Gerze Basin (KTc, PNT, PND, PNH, CJD, QT)									27.3 ± 4.5°N
A3 mean paleolatitude taking location of the Gerze Basin (PND, PNH, CJD)									24.5 ± 4.2°N
A4 mean paleolatitude taking location of the Gerze Basin (KTc, QT)									26.9 ± 4.7°N

(*continued on next page*)

Table 2 (*continued*)

Pole	Lithology	Slat/Slon (°N/°E)	Age (Ma)	Plat/Plon (°N/°E)	A_{95} (°)	P_λ (°N)	n (N)	Criterion (Q)	Reference
Indian craton, Europe and Asia									
IND			30 Ma	79.1/266.1	4.0	21.3			Besse and Courtillot (2002)
EUR			30 Ma	82.8/158.1	3.8	33.9			Besse and Courtillot (2002)
Asia			30 Ma	83.9/259.6	2.8	26.1			Cogné et al. (2013)

Abbreviations: volc, volcanic rocks; sed, sedimentary rocks; E_1, early paleogene; E_2, middle paleogene; Slat/Slon, latitude/longitude of sites; Plat/Plon, latitude/longitude of poles; A_{95}, the radius of circle of 95% confidence about the pole; P_λ, paleolatitude calculated for the reference point (32.2°N, 84.3°E); n (N), number of samples (sites) used to calculate the mean direction; Criterion (Q): data quality criteria (number of criteria met) after Van der Voo (1990) [1, well determined rock age; 2, sufficient number of samples, $n > 24$, $k \geqslant 10$, $\alpha_{95} \leqslant 16.0°$; 3, step-wise demagnetization; 4, field tests; 5, structural control and tectonic coherence with the craton discussed; 6, presence of reversal; 7, no resemblance to paleopole of younger age]; numbers mean satisfying the corresponding criterion, Q, number of the criteria satisfied; for criterion 4, F means positive fold test; for criterion 6, R means positive reversals test, D means dual-polarity; "□", failed to meet this criterion.
KTo*: the data before E/I corrections are not recommend for tectonic interpretation in this study.

Table 3
Selected Cenozoic paleomagnetic poles of the Siberia, Amuria, North China and South China terranes.

Pole	Lithology	Age (Ma)	Slat/Slon (°N/°E)	Plat/Plon (°N/°E)	dp/dm (A_{95}) (°)	P_λ (°N)	Reference
Siberia							
01	volc	19.9 ± 0.2	52.1/100.3	69.8/186.5	8.4/10.2	26.0	Hankard et al. (2007)
02	sed	19.5 ± 3.5	60.0/83.0	70.0/230.0	4.7/6.1	15.2	Gorbunov (1971)
Amuria							
03	volc	12.7 ± 0.3	45.5/101.0	71.6/178.0	14.6/18.2	29.2	Hankard et al. (2007)
04	volc	29.5 ± 6.5	43.5/104.5	70.0/204.0	8.3/11.5	20.9	Gorshkov et al. (1991)
05	volc	29.8 ± 1.8	45.4/101.3	81.9/275.6	3.5/4.8	24.2	Hankard et al. (2007)
06	volc	39.4 ± 0.6	44.5/101.4	72.0/202.6	6.3/8.5	22.5	Hankard et al. (2007)
07	volc	51.5 ± 1.5	44.0/135.0	78.4/236.7	10.3	21.8	Otofuji et al. (1995)
08	volc	57.1 ± 0.8	43.2/104.6	69.6/148.0	6.3/7.3	39.1	Hankard et al. (2008)
North China Block							
09	volc	15.0 ± 11.0	41.0/114.7	83.4/192.8	6.2	29.9	Zheng et al. (1991)
10	volc	16.0 ± 2.0	36.3/118.6	85.2/238.4	5.6	27.9	Zhao et al. (1994)
11	volc	16.0 ± 2.0	36.4/118.8	86.6/236.1	4.9	29.2	Zhao (1987)
12	volc	19.5 ± 4.5	40.2/112.7	83.7/232.3	5.4/7.5	26.8	Pan et al. (2005)
13	volc	20.0 ± 6.0	41.0/116.2	88.6/29.2	8.2	31.0	Zhao et al. (1994)
14	volc	20.0 ± 6.0	41.5/116.0	88.6/29.2	8.2	31.0	Zhao et al. (1990)
15	sed	14.0 ± 9.0	39.1/96.7	73.0/253.5	4.2/7.0	15.5	Dupont-Nivet and Butler (2003)
16	sed	28.5 ± 5.5	40.0/97.7	66.8/256.5	3.9/6.8	9.2	Dupont-Nivet and Butler (2003)
South China Block							
17	volc	13.0 ± 3.0	23.5/121.4	79.5/293.3	9.4/17.4	22.9	Hsu et al. (1966)
18	volc	14.0 ± 9.0	24.8/121.3	81.7/294.0	8.0/14.0	24.9	Hsu et al. (1966)
19	volc	15.5 ± 7.5	24.9/121.4	84.5/191.1	10.9/17.5	30.5	Miki et al. (1993)
20	sed	14.0 ± 9.0	30.5/111.5	74.7/40.8	11.2	42.5	Kent et al. (1986)
21	sed	44.0 ± 6.0	22.8/108.4	83.8/236.0	4.3	26.7	Zhao et al. (1994)
22	sed	47.0 ± 13.0	23.6/107.0	58.6/18.2	7.4	39.3	Gilder et al. (1993)
23	sed	49.5 ± 15.5	26.1/101.7	70.1/224.6	4.9	16.2	Yoshioka et al. (2003)
24	sed	49.5 ± 15.5	26.4/102.3	70.6/286.1	11.6	14.0	Huang and Opdyke (1992)
25	sed	60.0 ± 5.0	26.9/112.9	82.6/300.7	4.4	26.1	Sun et al. (2006)
26	sed	60.5 ± 4.5	25.7/101.3	72.3/218.4	4.5	19.2	Yoshioka et al. (2003)

Abbreviations: volc, volcanic rocks; sed, sedimentary rocks; Slat/Slon, latitude/longitude of sites; Plat/Plon, latitude/longitude of poles; A_{95}, radius of the circle of 95% confidence about the paleopole; dp/dm, semi-axes of elliptical error around the pole at a probability of 95%; P_λ, paleolatitude calculated for the reference point at 32.2°N, 84.3°E.

those of the second group, show a higher paleolatitude of 27.3 ± 4.5°N (A2 in Table 2).

In the first group, the inclinations of sedimentary rocks in the Fenghuoshan Group (including FHSL, FHSH, YPGX, FS, and SP in Table 2), the Yaxicuo Fm (YXC in Table 2) and the Nianbo Fm (NPC in Table 2) might have been flattened (Lippert et al., 2014). Regarding the poles DZ and NPS, from volcanic rocks of the Dianzhong Fm and Nianbo Fm respectively, they have not averaged out the paleosecular variations.

The second-group data, however, are considered more acceptable by some authors (Lippert et al., 2014), except for one high latitudinal anomaly (PNT in Fig. 10a, Table 2) where secular variations were not averaged out (Liebke et al., 2010; Sun et al., 2010; Lippert et al., 2014). The results from Cuojiangding Group (CJD in Fig. 10a, Table 2) passed a fold test (Meng et al., 2012). Dupont-nivet et al. (2010a) obtained a paleolatitude result of 23.2 ± 5.2°N from the Pana Fm (57–43 Ma) which passed both reversal and fold test. Lippert et al. (2014) recently reviewed published paleomagnetic data from the upper volcanic units of the Linzizong Fm and present a paleomagnetic pole that predicts the Gerze Basin would be located at 23.9 ± 4.0°N at ~50 Ma (PNL in Table 2). Huang et al. (2013) reported a result from the upper Pana Fm also, in which inclination was corrected using the E/I method and the AARM techniques (PNH in Fig. 10a, Table 2). The result agrees with that from its lower volcanic rocks (PND in Fig. 10a, Table 2), indicating that the northern margin of the Lhasa terrane was situated at about 24.0 ± 2.6°N during that time. The data of the Qiangtang terrane reported by Lippert et al. (2011) from volcanic rocks (34–38 Ma) in the Ejiumaima, Nading Co, and Ulan Ul lake areas (QT in Fig. 10a, Table 2) averaged out the secular variation, and possessed dual-polarity or passed reversal test, predict that the northern margin of the Lhasa terrane (reference point) was at about 27.9 ± 3.7°N.

Based on the above discussion, it is likely that inclination flattening or secular variation were the reasons for a lower paleolatitude of the Cenozoic paleomagnetic data of the Lhasa and

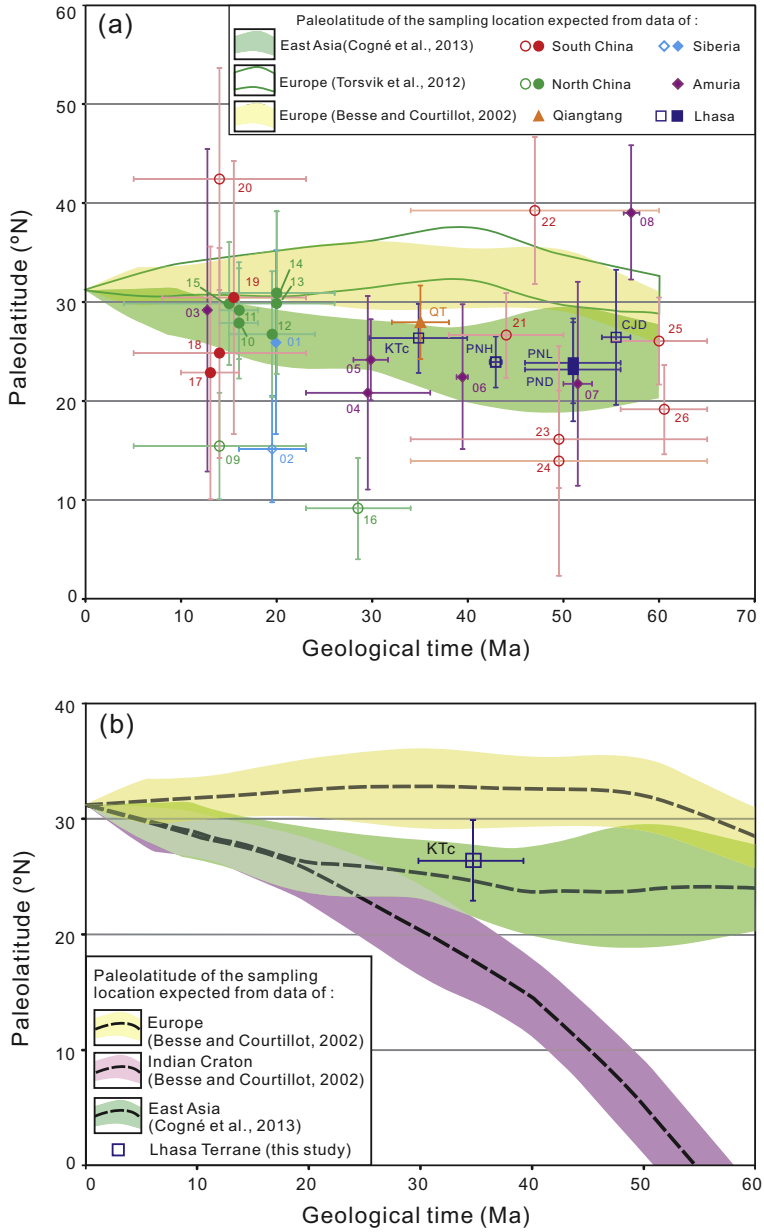


Fig. 10. (a) Paleolatitude plots for the Europe, East Asia, Siberia, and Amuria, North China, South China, Lhasa and Qiangtang terranes since 60 Ma. The expected and observed paleolatitudes were calculated for the reference point at 32.2°N, 84.3°E. For abbreviation and number of a pole see in Tables 2 and 3. Solid (open) symbols represent the data from volcanic (sedimentary) rocks. (b) Paleolatitude plots for the Europe, India and East Asia since 60 Ma, and Lhasa terrane at ~35 Ma.

Qiangtang terranes. The northern border of the Lhasa terrane should be at around 24.5 ± 4.2°N (A3 in Table 2) during 57–42 Ma as proposed by Lippert et al. (2014), and ~26.9 ± 4.7°N (A4 in Table 2) during 40–30 Ma (Fig. 10a). The similarity between paleomagnetic results from the Qiangtang and Lhasa terranes indicates that the convergence between these terranes has been small for the past 50 Ma and that the amount of corresponding shortening is small enough that it cannot be resolved paleomagnetically. Geological evidence (Kapp et al., 2005, 2007a, 2007b) also supports this viewpoint.

Before the inclination shallowing corrections, however, the paleomagnetic results of the Kangtuo Fm are these: D_s = 340.3°, I_s = 27.4°, k = 58.5, α_{95} = 3.2° and the paleomagnetic pole is at (64.8°N, 314.3°E, A_{95} = 2.6°), paleolatitude of the sampling location is 14.5 ± 2.6°N. But the coeval paleomagnetic data from the Indian craton (Besse and Courtillot, 2002) suggest a paleolatitude of 21.3 ± 4.0°N for our sampling location (Fig. 10b). They are obviously conflicting results. The inclination corrections of the samples of the Kangtuo Fm provided one possibility to solve this conflict.

Calculated from the ~30 Ma reference pole of the Europe APWP (Besse and Courtillot, 2002), the paleolatitude of the reference point (32.2°N, 84.3°E) in Gerze region is ~33.9 ± 3.8°N, which is 8.0 ± 4.9° greater than 25.9 ± 3.1°N observed by this study. This seems that there has been a ~880 ± 539 km shortening between the Lhasa terrane and the stable Europe since ~30 Ma. However, comparing with the data from the rest of Asia (selected by Cogné et al., 2013), our results indicate that there are no significant paleolatitudinal difference between the Lhasa terrane and other Asian terranes, e.g. the Qiangtang terrane, the South and North China terranes and even those terranes in Mongolia and Siberia as proposed by Lippert et al. (2011) (Fig. 10a), at ~30 Ma. Our study thus

reinforces the conclusion that there is a Cenozoic low paleolatitude anomaly in the Tibetan and eastern Asian regions.

Our data also permit us to estimate the amount of post-30 Ma convergence between the Lhasa terrane and India to the south. Using the ~30 Ma paleomagnetic pole from the India APWP (Besse and Courtillot, 2002), an expected paleolatitude of 21.3 ± 4.0°N is resulted for the Gerze region, which is 4.6 ± 5.1° lower than the 25.9 ± 3.1°N observed in this study, reflecting a ~506 ± 561 km shortening between the India Craton and the Lhasa terrane since the Oligocene (Fig. 10b). It is inferred that this amount of shortening has been absorbed mainly by the Himalaya Orogenic Belt. In the Himalaya Orogenic Belt (Fig. 1), the Main Central Thrust actuated at 23 Ma (Parrish and Hodges, 1996; Godin et al., 2001) and it moved southward by at least 75–100 km (Schelling and Arita, 1991), and the Main Boundary Thrust appeared at 12 Ma (Meigs et al., 1995; Huyghe et al., 2001), having displaced for about 100 km (Burbank et al., 1996). The Low Himalaya Fold-Thrust Zone was mobilized mainly after the MCT (DeCelles et al., 2001), where a ~200 km shortening has occurred (Mitra et al., 2010; Long et al., 2011). In southern Lhasa terrane, the mobilization period of the Gangdese Thrust-Fault System was constrained at 30–24 Ma (Harrison et al., 1999; Yin et al., 1999), during which at least 50 km shorting occurred (Yin et al., 1999; Murphy and Yin, 2003). The Greater Counter Thrust started working at 20 Ma (Yin et al., 1999; Ratschbacher et al., 1994), and the amount of shortening was at least 38 km (Murphy et al., 2009). The above-mentioned shortening of about 500 km is quite consistent with the paleomagnetic estimation (DeCelles et al., 2002; Johnson, 2002; Long et al., 2011).

5. Conclusions

The weak-field magnetic fabric of sedimentary rocks of the Oligocene Kangtuo Fm in Central Tibet shows no obvious deformation associated with tectonic stress after deposition; the fabric, however, is consistent with sediment particle settling and compaction. Demagnetization of the Kangtuo Fm specimens revealed two magnetic components: a normal polarity low temperature component that we interpret as an overprint by the present day field; and a high temperature component of which the lower Beds (1–93) have normal polarity, and the upper Beds (94–131) have reversed polarity decays toward the origin of orthogonal vector plots and is distinct from the present day field. The distribution of these characteristic directions is elongated in the horizontal plane, which is a hallmark of inclination flattening. We used the elongation/inclination correction method to transform the distribution of measured directions to a distribution consistent with non-flatted directions. Our results suggest that the magnitude of inclination shallowing in these fluvial–lacustrine sediments varies with particle grain size, with up to ~18° of total shallowing. The corrected inclinations pass a Class B reversal test. A comparison of our new results with the APWPs of stable Europe and India revealed a shortening of ~880 ± 539 km between the Lhasa terrane and Europe and ~506 ± 561 km between the Lhasa terrane and the India Craton, since ~30 Ma. Our results also indicate no significant convergence between the northern Lhasa terrane and the Qiangtang and other Asian terranes to the north. Therefore, we conclude that the majority of India–Asia convergence since 30 Ma has been absorbed south of our study area, in the Himalaya.

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Appendix A. Supplementary material

Supplementary data associated with this article can be found, in the online version, at http://dx.doi.org/10.1016/j.jseaes.2014.10.006.

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